Reg. No									

St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore – 607001

B.C.A DEGREE EXAMINATION APRIL 2021

ASCA202T – STATISTICAL METHODS

Time : 3 hrs **Max Marks :75**

SECTION – A (10X2=20)

Answer **ALL** the Questions

1. Define arithmetic mean.
2. Define range.
3. List out the types of skewness.
4. Define kurtosis.
5. Define correlation
6. State the regression equation of y on x.
7. Define standard error.
8. Define F-test
9. Write down the test statistic for testing a single proportion. Under large sample.
10. State the formula for testing correlation coefficient.

SECTION – B (5X5=25)

Answer **ALL** the Questions

11. (a) What do you mean by measures of central tendency.

(Or)

(b) Calculate the mean deviation about mean from the following data. 18,20,12,14,19,22,26,16,19,24.

12. (a) Write short notes an

(i) Karl Pearson's coefficient of skewness.

(ii) Bowley's co efficient of skewness.

(Or)

(b) Briefly explain kurtosis.

13. (a) Write shortly on type of correlation

(Or)

(b) Calculate the rank correlation co efficient from the following data.

Statistics	1 2 3 4 5 6 7 8 9 10
Mathematics	1 4 2 5 3 9 7 10 6 8

14. (a) Explain students t – test for testing single mean.

(Or)

(b) Explain chi-square test.

15. (a) Write shortly on single mean test under large sample.

(Or)

(b) Explain the test of significance based on correlation co efficient under large sample.

SECTION - C (3X10=30)

Answer any **THREE** Questions

16. Calculate mean, median and mode from the following data.

CI	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	5	7	12	16	6	4

17. Calculate Bowley's co – efficient of skewness for the following data.

Weight(in kgs)	40-50	50-60	60-70	70-80	80-90	90-100
No. of Persons	18	55	30	44	26	12

18. Calculate the karl pearsons correlation coefficient from the following data.

X	1 3 5 8 9 10
Y	3 4 8 10 12 11

19. A filling machine is expected to fill 5 kg of powder into bags. A sample of 10 bags gave the weights 4.7, 4.9, 5.0, 5.1, 5.4, 5.2, 4.6, 5.1, 4.6, and 4.7. Test whether the machine is working properly ($t_{0.05}$ (9) d f = 2.262)
20. State the steps involved in testing of hypothesis.

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